

NITHISH KANNAN M

+91 78451 36609 | nithishkannanm1@gmail.com | Chennai, Tamil Nadu, India | [GitHub](#) | [LinkedIn](#) | [nithishkannanm.com](#)

PROJECTS

Ring Zero — Predictive Memory Orchestrator

Samsung EnnovateX Hackathon, Apr – May 2026

- Built a predictive memory orchestrator — a multi-modal next-context transformer paired with a PPO-trained cgroup v2 memory-tier policy — to avoid cold-starting local models on mobile while memory is under KV-cache pressure.
- Outperformed the Android App Standby Buckets baseline: 90% vs 86% warm rate and 192ms vs 217ms latency at a 400MB memory budget.

Modular Memory Unit (MMU) — Hybrid Retrieval RAG Framework

May – Sep 2026

- Built a hybrid retrieval RAG framework combining BGE-M3, FAISS, and BM25 with Reciprocal Rank Fusion; benchmarked on legal/compliance documents across 5 custom metrics including faithfulness and cross-document reasoning.
- Shipped a FastAPI backend (HybridRetriever, async/sync thread-pool pattern, SSE streaming).

ex_mem — Contradiction-Aware Shared Memory Server (MCP)

Jun – Jul 2026

- Designing an MCP-based memory server that treats changed facts as evolution with preserved history rather than overwrite, targeting a gap where all 22 systems evaluated on MemoryAgentBench's FactConsolidation benchmark score $\leq 7\%$.
- Architected a hybrid SQLite FTS5 + sqlite-vec retrieval stack with an LLM-judge-based contradiction classifier.

Plateau — Semantic Circuit Breaker for AI Agents

VIT Chennai Frontier AI Hackathon, Jul 2026

- Built a detector that trips on semantic information stagnation (not textual repetition) in autonomous agent loops, using offline sentence-embedding comparison with a self-calibrating baseline.
- Validated with 121 passing tests and a live two-machine demo; shipped as the installable package plateau.

TECHNICAL SKILLS

Languages: Python, C, C++, SQL

ML / Retrieval: PyTorch, LangChain (LCEL), BGE-M3, FAISS, BM25, Hybrid Retrieval (RRF Fusion)

Systems & Infra: Linux cgroup v2 & PSI telemetry, Docker, Kubernetes, OpenTelemetry, MCP (Model Context Protocol), AWS

Web & Tools: FastAPI, React, HTML, CSS, JavaScript, Git/GitHub

EXPERIENCE

Deep Learning Engineer — AI Club, VIT Chennai

Jan 2026 – Present

- Build and review deep learning systems and research-oriented prototypes as part of the club's core AI engineering team.
- Delivered a speaker session on Cursor, an AI coding assistant, as part of an AI Club workshop.

AI/ML Engineer — AWS Cloud Club, VIT Chennai

Feb 2026 – Present

- Develop ML and cloud-integrated engineering projects on AWS infrastructure within the AWS Student Builder community.

Freelance Web/Infra Consultant — Viha Online

June 2026– Present

- Built a diagnostic pipeline against Google Search Console and Merchant Center APIs (via Google Cloud Console) alongside Shopify's Admin GraphQL API to cross-reference checkout, traffic, and search-ranking data, surfacing a 58% checkout abandonment rate and a 31.5-vs-10.8 desktop/mobile search-position gap.
- Traced a store-wide AOV drop to a single underperforming SKU by chaining automated ShopifyQL queries through the same pipeline, isolating a 7-8x landing-page traffic gap the client's team hadn't caught manually.
- Scoped a Carthike-based WhatsApp automation workflow and a proposed voice-AI customer care pipeline (Exotel telephony, Sarvam AI, Claude Haiku, ElevenLabs) for COD confirmation and order-status handling, pending client sign-off.

EDUCATION

Vellore Institute of Technology (VIT), Chennai

2024 – 2028

B.Tech, Computer Science and Engineering | CGPA: 8.34/10

CERTIFICATIONS & ACHIEVEMENTS

- Machine Learning Specialization — Stanford University / DeepLearning.AI · Generative AI Engineering — IBM.
- 3rd Place, BIS National Hackathon 2025 — Empath, an emotion-aware adaptive learning platform (React, FastAPI, HuggingFace, Supabase).
- GlucoSentry — Patent application under review (VIT IPR Cell) for an ESP32 wearable design with signal-quality-gated ECG acquisition, CNN+BiLSTM hypoglycemia risk inference, and an adaptive sampling/alert controller.